

Exercises

I) Theory: Derive the following parts:

1. [The Backpropagation algorithm](#)
2. [The derivative of sigmoid function](#)

II) Coding: Following the content and codes of chapter 12 of Python Machine Learning book, the dataset is of course MNIST

1. Compare the accuracy results between 20 and 40 hidden neuron network, using the same 1,000 training images, cross-entropy cost function, learning rate of $\eta=0.5$, mini-batch size of 10, and 300 epochs.
2. With a 40 hidden neuron network, compare the accuracy results between 1,000 and 5,000 training images, using cross-entropy cost function, learning rate of $\eta=0.5$, mini-batch size of 10, and 300 epochs.
3. With a 40 hidden neuron network, compare the accuracy results between with and without regularization, using the same 1,000 training images, cross-entropy cost function, learning rate of $\eta=0.5$, mini-batch size of 10, and 300 epochs, λ is chosen in $\{0.1, 1, 10\}$
4. Using heuristic approach, find the best hyper-parameters for learning a 40 hidden neuron network using the same 3,000 training images, cross-entropy cost function, mini-batch size of 10, and 300 epochs. Show the accuracy of learning epochs with these chosen parameters.

III) What is a CNN? Can you elaborate on each main component of a CNN? If the input image is 32×32 , the kernel size is 3×3 , the stride is 1, and zero padding, what is the output feature map size?

IV) Why is pooling layer important in CNN? What happens if stride increases? Why do CNNs use fewer parameters than fully connected networks? What is the difference between padding="valid" and padding="same"?

V) Build a simple CNN by using TensorFlow or PyTorch. Train a CNN on the MNIST dataset.

VI) Without using `Conv2D` or `nn.Conv2d`, implement convolution manually using matrix multiplication.